

DMMST: Experimental Results

Dynamic Multi-modal Multi-event Survival Transformer — complete results report

October 2, 2026

Abstract

This report collects every experiment run for the paper: two single-event benchmarks (METABRIC, SUPPORT), three multi-event datasets (a simulator with known truth, hsa_synthetic, EBMT), simulator sweeps over eight properties of the data, an evaluation of within-patient event ordering against true event times, tuning of the loss penalties, a continuous vs. quantile vs. discretised input comparison, three generations of experiments on coded visit histories (§4.2 of the paper) and the end-to-end language model (§3). The transformer trained with the piecewise-constant-hazard likelihood ($\mathcal{L}_{\text{PCH}}$) is the best or tied-best of the models we ran on every benchmark dataset, is the most accurate learned model at ordering events within a patient, and needs about 5000 patients to beat Cox; it remains below the published SurvTRACE. The ranking penalties $\mathcal{L}_{\text{rank}}$ and $\mathcal{L}_{\text{mul}}$ never improve a proper score, and the within-subject C-index is shown to be gameable. The coded-sequence claim (§4.2) is not supported by our experiments, and the end-to-end language model works but is clearly weaker.

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Summary of findings

1. **The base model is strong.** Our transformer with $\mathcal{L}_{\text{PCH}}$ has the best or tied-best C_{td} and the best calibration among neural models we ran on METABRIC (0.675), SUPPORT (0.619), hsa_synthetic (0.867), EBMT (0.545; all models 0.50–0.55) and the simulator (0.713), and matches the published PC-Hazard, DeepHit and Cox values (Section 2).
2. **It remains below the published SurvTRACE:** 0.675 vs. 0.691 (METABRIC) and 0.619 vs. 0.640 (SUPPORT).
3. **It orders events within a patient best among learned models**, measured against the simulator’s true event times; DeepHit, a competing-risk model, ranks patients well but orders events badly (Section 4). This supports the paper’s general multi-event formulation.
4. **$\mathcal{L}_{\text{rank}}$ and $\mathcal{L}_{\text{mul}}$ do not help.** Untuned they lower calibration and true ordering; tuned on validation they become indistinguishable from $\mathcal{L}_{\text{PCH}}$ (Section 4.1).
5. **The within-subject C-index can be gamed.** $\mathcal{L}_{\text{mul}}$ raises it above the oracle while every true-order score gets worse (Section 4).
6. **When a neural model pays off.** Cox matches the oracle for linear risk and collapses for non-linear risk; from ≈ 5000 patients the transformer leads clearly; it scales to 8 events (Section 5).
7. **MMV cuts the error of predicted event times on observed patients** (SUPPORT -21% , METABRIC -22%) at a cost in calibration (Section 2).
8. **Numeric encoding.** With correct preprocessing, continuous and quantile encodings are equal and both beat 10-bin discretisation on METABRIC; on SUPPORT all three are equal (Section 6).
9. **Coded sequences (§4.2) are not supported.** Even when the outcome depends on order and trends that a bag-of-codes table cannot see, no sequence model — including a larger model with [CLS] pooling and recency encoding — beats the table (Section 7).
10. **The end-to-end LLM (§3) works end to end but trails our transformer by 0.06–0.09 C_{td}** (Section 8).

1 Setup

1.1 Datasets

Table 1: Datasets. “Regime”: which events can be observed together.

Dataset	n	events	regime	features	time unit
METABRIC	1 904	1	single event	9 (5 numeric, 4 categorical)	months
SUPPORT	8 873	1	single event	14 (8 numeric, 6 categorical)	days
hsa_synthetic	5 000	2	non-competing	15 numeric	days
EBMT (bone-marrow transplant)	2 279	5	semi-competing (death censors others)	4 categorical	days
Simulator (sim_*)	500–10 000	2–8	any (default non-competing)	15 numeric, true CIFs known	arbitrary
EHR simulator v1 / v2	5 000	4	semi-competing	age, sex + coded visit history (≈ 70 –80 codes)	days

- **Simulator** (sat/data/simulate.py): Weibull proportional hazards per event with a risk that mixes shared and event-specific parts, optional non-linearity, unobserved frailty, per-event timescales (“order spread”) and calibrated censoring, in the competing, semi-competing or non-competing regime. Its true cumulative incidence functions (CIFs), verified against Monte Carlo, give an *oracle* row.
- **EHR simulator** (sat/data/simulate_ehr.py): five years of visits with diagnosis codes (hypertension, diabetes, CKD, COPD), medications, labs with values (eGFR, HbA1c, FEV₁), exacerbations and noise codes; four outcomes (CKD progression, diabetic complication, COPD

hospitalisation, death). A “temporal share” controls how much risk depends on order, trends and recency rather than current levels. In v1 the trend information was recoverable from counts and last values; v2 removes that shortcut.

1.2 Models

- **Ours**: numeric-value-embedding transformer (§2.2, Fig. 2 of the paper) → shared MLP → piecewise-constant-hazard head. Loss variants: $\mathcal{L}_{\text{PCH}}$; $+\mathcal{L}_{\text{rank}}$ (across patients, Eq. 4); $+\mathcal{L}_{\text{mul}}$ (within patient, Eq. 5); the paper recipe $\mathcal{L}_{\text{PCH}}+\mathcal{L}_{\text{rank}}+\mathcal{L}_{\text{mul}}$; $+\text{MMV}$ (UniSurv mean/variance loss with weight λ_v); “full” = all four.
- **Baselines**: cause-specific Cox PH; DeepHit, DSM and MENSA as published, sharing our transformer backbone and using their own heads and losses; the oracle on simulated data.
- **End-to-end LLM** (§3): DistilGPT2 (pretrained) or a 4-layer GPT-2 trained from scratch that reads the record as text and answers one token per time interval ($\mathbf{N}$, $\mathbf{C}$, $\mathbf{E}k$); per-event curves are derived from the token probabilities.

1.3 Metrics

$C_{td} \uparrow$	Time-dependent concordance across patients (IPCW, τ -truncated, as in SurvTRACE) at the 25/50/75 % quantiles of observed event times; multi-event data use each event’s own quantiles; averaged over horizons and events.
Brier $\downarrow$	Calibration at the same horizons.
Within-subject $C \uparrow$	Eq. 5 pairs: at the time an event happened, was it ranked more likely than the patient’s later or censored events? Shown to be gameable (Section 4).
True-order C / Brier / log-loss	Simulated data only: predicted $P(T_a < T_b \mid x)$, computed from each model’s per-event curves, scored against the true latent event times. Brier and log-loss are proper scores; no model can beat the oracle on them in expectation.
MAE (uncensored / margin)	Error of the predicted event time; “margin” also scores censored patients via a Kaplan–Meier best guess (unreliable on EBMT, see Section 10).

1.4 Protocol

Seed s sets both the initialisation and the train/validation/test split s (60/10/30 %); single-event and multi-event benchmarks use 5 seeds, sweeps and the LLM 3. Checkpoints are selected on validation C_{td} with early stopping. Tables report mean $\pm$ sd over seeds. “Paired” comparisons use the same seed for both models and report the mean difference, the number of seeds on which the difference is positive, and a paired t -test p -value (with 5 seeds the smallest Wilcoxon p is 0.0625, so it is not used). All runs were executed on Kaggle (T4 GPU) from the public repository; hyper-parameters are listed in Appendix A, and the corrections made before the reported runs in Appendix B.

2 Single-event benchmarks: METABRIC and SUPPORT

Table 2: C_{td} at the 25/50/75 % horizons, 5 splits, next to SurvTRACE (Wang & Sun, 2022) Table 2 (10 runs on their own splits). SUPPORT is the corrected run (notebook 12, Appendix B). Backbones tuned per dataset on validation.

Model	METABRIC				SUPPORT			
	25 %	50 %	75 %	mean	25 %	50 %	75 %	mean
<i>Ours (5 splits)</i>								
Cox PH	0.629	0.616	0.630	0.625	0.544	0.559	0.584	0.562
DeepHit	0.697	0.667	0.645	0.670	0.637	0.601	0.591	0.610
DSM	0.678	0.644	0.626	0.649	0.600	0.577	0.572	0.583
MENSA	0.681	0.648	0.621	0.650	0.557	0.532	0.521	0.536
Ours $\mathcal{L}_{PCH}$	0.700	0.675	0.649	0.675	0.650	0.610	0.599	0.619
<i>Published (SurvTRACE Table 2)</i>								
Cox PH	0.628	0.627	0.632	0.629	0.549	0.564	0.586	0.566
DeepHit	0.712	0.657	0.603	0.657	0.650	0.602	0.574	0.609
DSM	0.707	0.663	0.636	0.669	0.640	0.609	0.596	0.615
PC-Hazard	0.713	0.680	0.644	0.679	0.652	0.620	0.607	0.626
SurvTRACE w/o multi-task	0.722	0.686	0.649	0.686	0.665	0.630	0.614	0.636
SurvTRACE	0.728	0.690	0.655	0.691	0.670	0.633	0.617	0.640

Reading Table 2. Our reproduced baselines match their published values closely: Cox (0.625 vs. 0.629; 0.562 vs. 0.566) and DeepHit on SUPPORT (0.610 vs. 0.609); DeepHit on METABRIC is higher than published (0.670 vs. 0.657). This validates the evaluation pipeline. Our $\mathcal{L}_{PCH}$ model is the best of the models we ran on both datasets and at every horizon, and is at the level of the published PC-Hazard (0.675 vs. 0.679; 0.619 vs. 0.626). The published SurvTRACE remains 0.016 (METABRIC) and 0.021 (SUPPORT) ahead; its multi-task head accounts for 0.004–0.005 of that in its own ablation. DSM is unstable across splits (sd 0.048–0.049) and MENSA is weak on SUPPORT (0.536), as expected for a multi-event model on single-event data.

Table 3: Loss ablation for our model: C_{td} , Brier and event-time error (METABRIC: months; SUPPORT: days), 5 splits. “full” = $\mathcal{L}_{PCH} + \mathcal{L}_{rank} + \mathcal{L}_{mul} + \text{MMV}$ ($\mathcal{L}_{mul}$ is inactive with one event).

Loss	METABRIC				SUPPORT			
	C_{td}	Brier	MAE-unc.	MAE-margin	C_{td}	Brier	MAE-unc.	MAE-margin
$\mathcal{L}_{PCH}$	0.675 $\pm$ 0.014	0.177	69.6	80.4	0.619 $\pm$ 0.008	0.197	591	793
+ $\mathcal{L}_{rank}$	0.664 $\pm$ 0.008	0.185	66.8	84.3	0.619 $\pm$ 0.006	0.203	572	812
+MMV ($\lambda_v=0.01$)	0.666 $\pm$ 0.011	0.190	59.0	81.3	0.608 $\pm$ 0.008	0.199	568	780
full, $\lambda_v=0.01$	0.658 $\pm$ 0.015	0.196	57.4	83.7	0.613 $\pm$ 0.008	0.204	555	803
full, $\lambda_v=0.1$	0.667 $\pm$ 0.012	0.189	56.6	84.0	0.612 $\pm$ 0.005	0.203	546	800
full, $\lambda_v=1$	0.669 $\pm$ 0.012	0.244	54.1	91.8	0.616 $\pm$ 0.011	0.204	469	768
DeepHit (reference)	0.670 $\pm$ 0.013	0.189	75.4	86.5	0.610 $\pm$ 0.003	0.216	543	802

Reading Table 3. $\mathcal{L}_{rank}$ never improves C_{td} (METABRIC -0.010 , 0/5 seeds positive, $p = 0.05$; SUPPORT -0.0003 , $p = 0.91$) and worsens Brier in 5/5 splits on both datasets. MMV does what it is designed for: at $\lambda_v = 1$ the error on observed patients falls from 69.6 to 54.1 months on METABRIC and from 591 to 469 days on SUPPORT (-122 days, 5/5, $p < 10^{-4}$), with C_{td} unchanged ($p = 0.47$ on SUPPORT). The cost is calibration: Brier rises by 0.067 on METABRIC and by 0.007 on SUPPORT. The censoring-aware MAE-margin improves on SUPPORT (-25 days, 5/5, $p = 0.007$) but worsens on METABRIC (+11 months). MMV is therefore useful when the predicted time itself matters, not for ranking or calibration.

3 Multi-event benchmarks

Table 4: Multi-event datasets, 5 splits (three EBMT runs that crashed were re-run separately and agree with the table). Within-C = within-subject C-index; KM = the same score from population Kaplan–Meier curves (no patient information).

Model	simulator, base (4 events)			hsa (2 events)	EBMT (5 events)	
	C_{td}	Brier	Within-C	C_{td}	C_{td}	Within-C
Oracle	0.736	0.156	0.696	–	–	–
Population KM	–	–	0.580	–	–	0.837
Cox PH	0.679	0.175	0.647	0.492	0.533	0.843
DeepHit	0.712	0.230	0.678	0.865	0.528	0.841
DSM	0.712	0.167	0.668	0.865	0.501	0.189
MENSA	0.701	0.172	0.649	0.865	0.504	0.465
Ours $\mathcal{L}_{PCH}$	0.713	0.166	0.674	0.867	0.545	0.840
Ours $+\mathcal{L}_{mul}$	0.700	0.172	0.666	0.846	0.535	0.880
Ours $+\mathcal{L}_{rank}$	0.713	0.180	0.639	0.868	0.543	0.838
Ours paper recipe	0.713	0.182	0.658	0.864	0.537	0.879
Ours full ($\lambda_v=1$)	0.712	0.213	0.673	0.858	0.550	0.894

Reading Table 4. On the simulator our $\mathcal{L}_{PCH}$ model is the closest learned model to the oracle (C_{td} 0.713 vs. 0.736) with the best Brier; DeepHit and DSM tie on C_{td} but are worse calibrated. On hsa_synthetic the signal is entirely non-linear: Cox is at chance (0.49) while every neural model reaches ≈ 0.865 . EBMT’s four categorical covariates carry little signal (C_{td} 0.50–0.55 for every model), and the population order of events is already highly predictable (KM within-C 0.837). Adding $\mathcal{L}_{mul}$ lowers C_{td} on the simulator (-0.013 , 0/5, $p = 0.002$) and on hsa (-0.021 , 0/5, $p < 0.001$). On EBMT it raises within-C from 0.84 to 0.88 — but EBMT is semi-competing, exactly the setting in which Section 4 shows this metric is gamed. DSM and MENSA order EBMT’s events worse than the population (0.19 and 0.47).

4 Within-patient event ordering against the true event times

Table 5: Ordering scored against the simulator’s true event times, 5 splits. TO = true order; TO-Brier and TO-log-loss are proper scores.

Model	non-competing (base)					semi-competing			
	C_{td}	Within-C	TO-C	TO-Brier	TO-log-loss	C_{td}	Within-C	TO-C	TO-Brier
Oracle	0.736	0.696	0.685	0.203	0.592	0.734	0.739	0.673	0.210
Cox PH	0.679	0.647	0.624	0.227	0.645	0.679	0.716	0.623	0.227
DeepHit	0.711	0.674	0.552	0.244	0.681	0.712	0.546	0.559	0.281
MENSA	0.709	0.664	0.653	0.218	0.626	0.708	0.715	0.644	0.221
Ours $\mathcal{L}_{PCH}$	0.714	0.673	0.659	0.215	0.618	0.714	0.738	0.658	0.214
Ours $+\mathcal{L}_{mul}$	0.709	0.671	0.657	0.220	0.630	0.703	0.787	0.633	0.228
Ours paper recipe	0.715	0.659	0.653	0.230	0.653	0.713	0.781	0.618	0.232

Red: above the oracle on the gameable within-C while worse than $\mathcal{L}_{PCH}$ on every true-order score.

Reading Table 5. Measured against the true event times, our $\mathcal{L}_{PCH}$ model is the best learned model in both regimes, ahead of MENSA, Cox and DeepHit. Adding $\mathcal{L}_{mul}$ or using the paper recipe makes every true-order score worse (TO-Brier: 0/5 improvements in all four comparisons, $p < 0.001$). In the semi-competing regime $\mathcal{L}_{mul}$ lifts the within-subject C-index to 0.787 — above the oracle’s 0.739 — while its true-order C falls by 0.025. The penalty is trained on the same pairs the metric counts and learns to reshape the event curves around the terminal event, which wins those pairs

without predicting better. DeepHit has a good C_{td} but the worst true ordering (0.552): its likelihood assumes one event per patient and is misspecified when events co-occur, which is the case for our general multi-event formulation.

4.1 Tuning the penalties

Table 6: Penalty weight and sharpness σ chosen on validation C_{td} (grid $\sigma \in \{0.1, 0.5, 1\}$, weight $\in \{0.01, 0.05, 0.2\}$, split 0), then tested on 5 splits.

Penalty (data)	chosen setting	model	C_{td}	Brier	TO-Brier
$\mathcal{L}_{\text{rank}}$ (METABRIC)	$\sigma=0.5$, weight 0.01	$\mathcal{L}_{\text{PCH}}$	0.6705	0.1810	–
		tuned $\mathcal{L}_{\text{rank}}$	0.6718	0.1776	–
$\mathcal{L}_{\text{mul}}$ (simulator)	$\sigma=0.5$, weight 0.2	$\mathcal{L}_{\text{PCH}}$	0.7152	0.1662	0.2148
		tuned $\mathcal{L}_{\text{mul}}$	0.7139	0.1661	0.2145

Reading Table 6. Both validation grids ranked $\sigma = 0.1$ — the value used in all untuned runs — last. At the best setting each penalty is indistinguishable from $\mathcal{L}_{\text{PCH}}$ (METABRIC: 3/5, $p = 0.81$). Tuning removes the harm but adds no benefit. Our interpretation: $\mathcal{L}_{\text{PCH}}$ is the full likelihood of the event times, so it already learns the ranking and the within-patient order the model can represent; a ranking penalty can only pull the predicted curves away from that optimum.

5 Simulator sweeps

Each figure varies one property of the simulated data around the base scenario (3 splits; error bars: sd). The dashed grey line is the oracle. Panels: within-subject C, C_{td} , within-C gain over population KM, and (sweeps of notebook 11 only) true-order Brier.

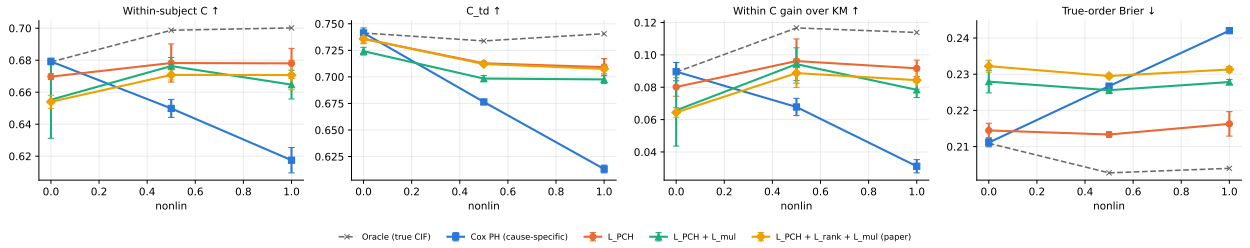


Figure 1: Non-linearity of the risk (0 = linear, 1 = fully non-linear).

Figure 1. For a linear risk Cox equals the oracle (C_{td} 0.741) and our model loses little (0.736). As the risk becomes non-linear Cox collapses (0.613) while ours stays at 0.709 (oracle 0.741). This is the clearest argument for a neural model.

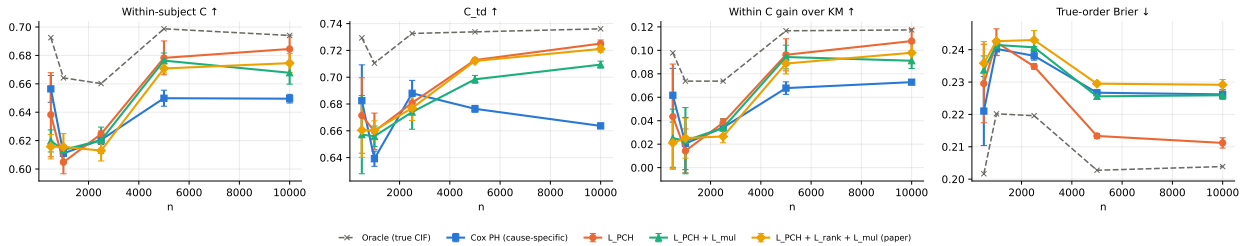


Figure 2: Training-set size (500 to 10000 subjects).

Figure 2. The gap of our model to the oracle shrinks from 0.058 ($n=500$) to 0.011 ($n=10\,000$); Cox’s gap stays 0.05–0.07. Below 5 000 subjects the two are comparable (Cox ahead at $n = 500$ and 2 500, the transformer at 1 000); from $\approx 5\,000$ subjects the transformer leads clearly. Each n is a different random dataset, so the small- n points are noisy.

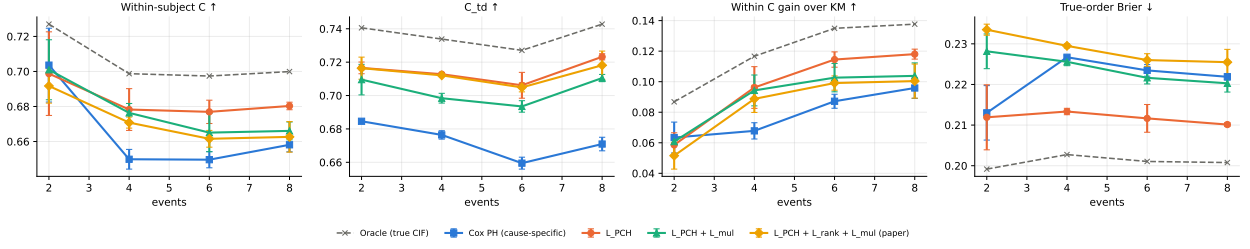


Figure 3: Number of events K (2 to 8).

Figure 3. Our model stays ≈ 0.02 below the oracle at every K , Cox 0.05–0.07 below: the framework scales with the number of events. $\mathcal{L}_{mul}$ and the paper recipe are worse than $\mathcal{L}_{PCH}$ on true-order Brier at every K (e.g. 0.226 and 0.230 vs. 0.213 at $K=4$).

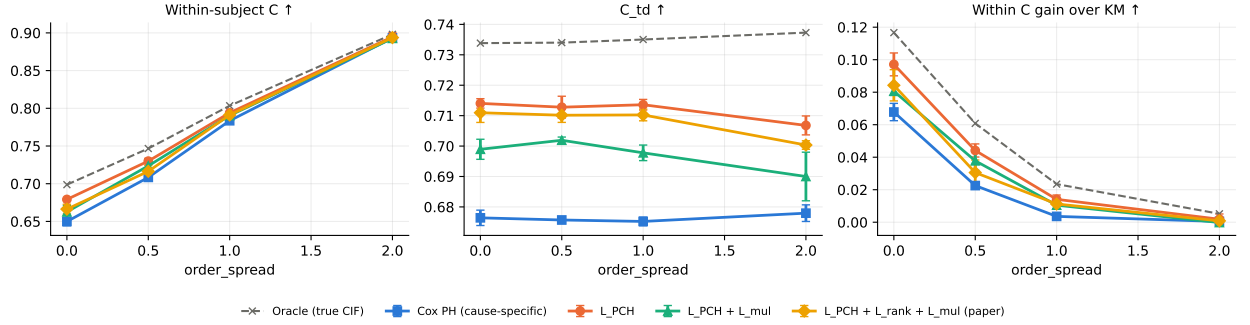


Figure 4: Spread of the events’ typical timing (0: the covariates decide the order; 2: nearly the same order for every patient).

Figure 4. When the order is the same for everybody (spread 2) every model reaches the oracle’s within-C (≈ 0.89 –0.90) and there is nothing patient-specific to learn. When the order is patient-specific (spread 0) the oracle reaches 0.699, $\mathcal{L}_{PCH}$ 0.679 and $\mathcal{L}_{mul}$ 0.663: $\mathcal{L}_{mul}$ is lowest exactly where it was meant to help.

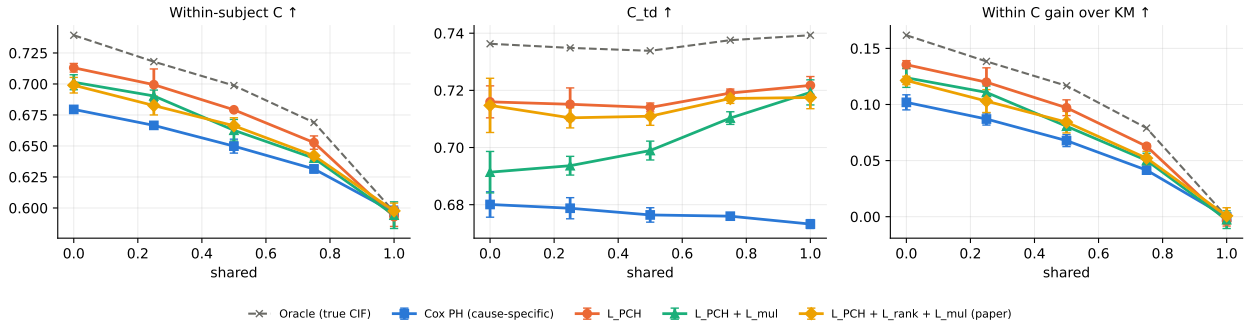


Figure 5: Share of the risk common to all events (“shared disease pathway”).

Figure 5. The more risk the events share, the less patient-specific ordering there is to learn; at a share of 1 even the oracle gains nothing over population ordering. Our model tracks the oracle at every level, and $\mathcal{L}_{mul}$ is never above $\mathcal{L}_{PCH}$.

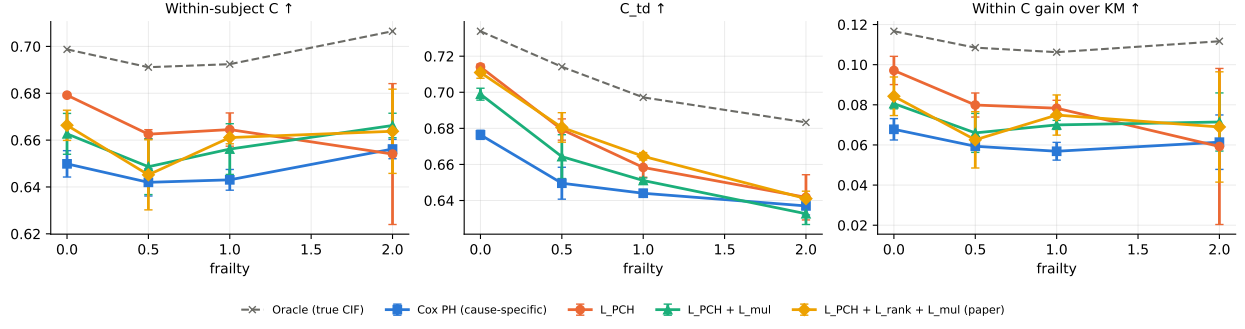


Figure 6: Unobserved shared frailty (dependence that the covariates cannot explain).

Figure 6. Frailty lowers every model and the oracle together ($\mathcal{L}_{\text{PCH}} C_{td}$ 0.714 $\rightarrow$ 0.642 at frailty variance 2); no loss variant is more robust than $\mathcal{L}_{\text{PCH}}$.

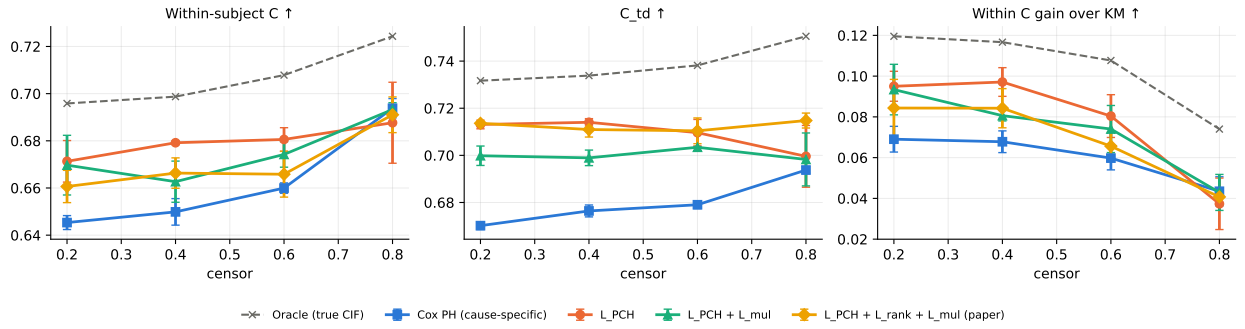


Figure 7: Fraction of unobserved event slots (censoring, 20–80 %).

Figure 7. Performance is stable across censoring levels ($\mathcal{L}_{\text{PCH}} C_{td}$ 0.700–0.714) and the ranking of the models does not change.

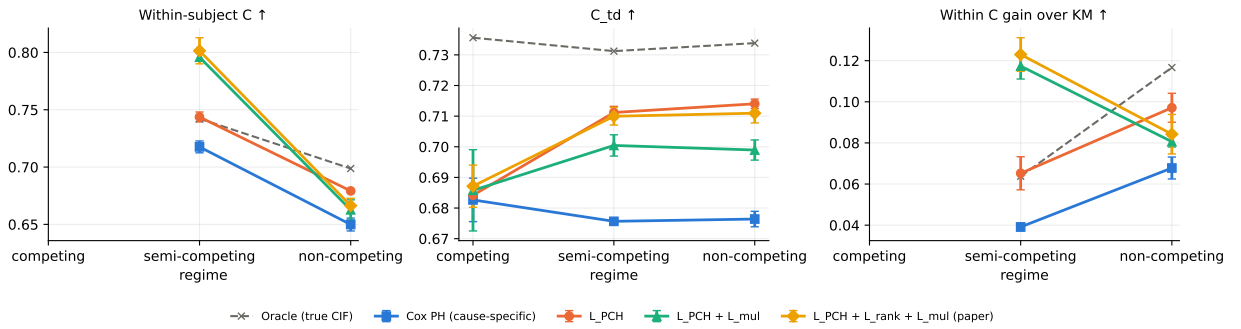


Figure 8: Observation regime: competing, semi-competing, non-competing.

Figure 8. Under competing risks only the first event is observed, so no within-patient pairs exist and within-C is undefined. Under semi-competing risks $\mathcal{L}_{\text{mul}}$ and the paper recipe reach within-C 0.796–0.802, far above the oracle’s 0.742 — the effect explained in Section 4. C_{td} of our model is similar across regimes (≈ 0.70 –0.71).

6 Numeric encoding: continuous, quantile and discretised input

Table 7: Same model, backbone and splits; numeric features standardised (continuous), rank-transformed to a normal distribution (quantile) or cut into 10 quantile bins (discretised). C_{td} , 5 splits (notebook 12, corrected preprocessing).

Model	METABRIC			SUPPORT		
	continuous	quantile	discretised	continuous	quantile	discretised
Ours $\mathcal{L}_{PCH}$	0.669 $\pm$ 0.008	0.669 $\pm$ 0.010	0.651 $\pm$ 0.019	0.618 $\pm$ 0.007	0.619 $\pm$ 0.006	0.621 $\pm$ 0.005
DeepHit	0.663 $\pm$ 0.018	0.655 $\pm$ 0.027	0.650 $\pm$ 0.020	0.607 $\pm$ 0.004	0.609 $\pm$ 0.005	0.615 $\pm$ 0.009

Reading Table 7. On METABRIC continuous and quantile encodings are equal (0.669) and both beat discretisation (-0.018 , $0/5$, $p = 0.03$ for our model). On SUPPORT, after the preprocessing correction, the three encodings are statistically indistinguishable for both models (all $p \geq 0.2$); the earlier apparent advantage of discretisation on SUPPORT was caused by the preprocessing issue (Appendix B). The paper can therefore state that the numeric-value embedding is at least as good as discretisation and better on METABRIC, but not that it always helps. Our METABRIC continuous value here (0.669) differs from Table 2 (0.675) only by run-to-run variation (sd 0.008–0.014).

7 Coded visit histories (§4.2)

Three generations of experiments test whether the transformer can use the order, trends and recency of a coded history: v1 (notebook 06), v2 (notebook 09, removes the shortcut through counts and last values) and the final runs on v2 with improved sequence models (notebook 13). In every case the same patients are given to the model as the full time-stamped sequence or as a bag-of-codes table (counts per code, last value per lab, number of visits).

Table 8: v1 and v2 cohorts with the small model (hidden 16, 2 layers, mean pooling). C_{td} , 3 splits. “temporal” = share of the risk carried by order, trends and recency.

Design	cohort (temporal)	Oracle	Cox [bag]	Ours [bag]	Ours [sequence]	MENSA [sequence]	Ours [static]
v1	static (0.0)	0.735	0.725	0.710	0.708	0.712	0.502
v1	base (0.5)	0.723	0.708	0.705	0.700	0.690	0.500
v1	temporal (0.9)	0.708	0.688	0.686	0.684	0.675	0.500
v2	level (0.0)	0.745	0.735	0.716	0.726	0.721	0.502
v2	mixed (0.5)	0.725	0.697	0.691	0.687	0.681	0.498
v2	order (0.9)	0.696	0.653	0.655	0.647	0.645	0.502

Table 9: Final runs on the v2 cohorts, 5 splits. “large” = hidden 64, 4 layers, [CLS] pooling, $\text{lr } 10^{-3}$; `sequence_time` adds the recency $\exp(t/1 \text{ year})$ to every code.

Model [input]	C_{td}			True-order Brier $\downarrow$		
	level (0.0)	mixed (0.5)	order (0.9)	level	mixed	order
Oracle	0.743	0.721	0.692	0.209	0.218	0.223
Cox PH [bag]	0.730	0.692	0.648	0.210	0.226	0.235
Ours small [bag]	0.711	0.686	0.648	0.221	0.229	0.235
Ours large [bag]	0.693	0.668	0.630	0.227	0.235	0.238
Ours small [sequence]	0.721	0.685	0.645	0.217	0.230	0.235
Ours large [sequence]	0.688	0.663	0.630	0.234	0.240	0.244
Ours large [sequence_time]	0.686	0.657	0.627	0.233	0.240	0.242

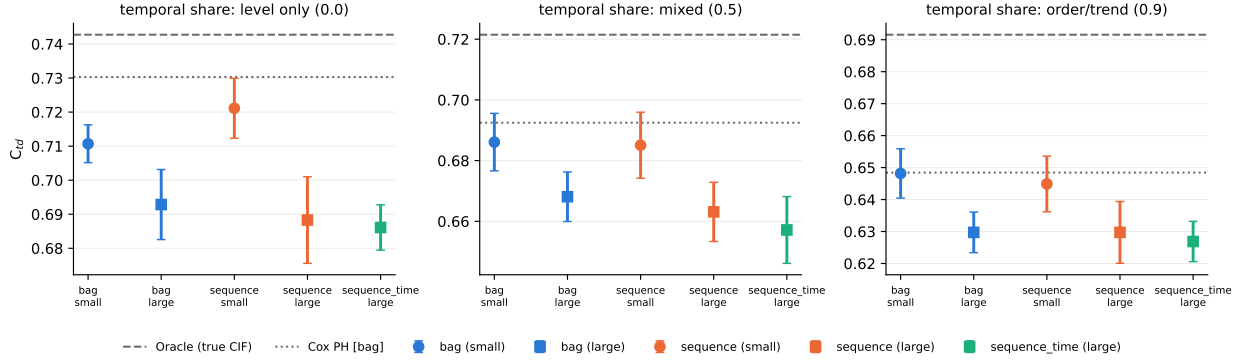


Figure 9: Final §4.2 runs (Table 9): C_{td} of each input/model variant, mean $\pm$ sd over 5 splits. Dashed: oracle; dotted: Cox on the bag table.

Reading Tables 8–9 and Figure 9.

- **v1:** sequence and bag tie because the trend information could be recovered from counts and last values (a linear model on the bag recovers 66–86 % of the true risk).
- **v2** removes this shortcut (the bag recovers only 12–41 % for three of the four outcomes), and the oracle’s lead over the best learned model grows with the temporal share: 0.013 (level), 0.029 (mixed), 0.044 (order). The information is present in the history.
- **No sequence model extracts it.** The small sequence model beats the bag only in the level-only cohort (+0.010, 5/5, $p = 0.046$), where no temporal information exists, and ties it elsewhere (-0.001 , -0.003 , $p \geq 0.09$).
- **The larger model is worse**, for both inputs and in all cohorts (-0.015 to -0.033 C_{td} , 0/5, $p \leq 0.04$), most likely overfitting ≈ 3000 training patients. Recency encoding adds nothing ($p \geq 0.18$).
- Cox on the bag table is the best learned model on these cohorts, consistent with their near-linear risk in the bag features.

Conclusion: under the conditions tested — 5000 patients, up to 256 tokens, small and medium transformers — the §4.2 claim that modelling the coded sequence improves prediction over a summary table is not supported.

8 End-to-end language model (§3)

Table 10: The §3 language model vs. Cox and our transformer on the same splits. C_{td} , 3 splits.

Model	simulator (base)	EBMT	METABRIC
Oracle	0.734	–	–
Cox PH	0.676	0.538	0.624
Ours $\mathcal{L}_{PCH}$	0.713	0.544	0.666
Ours paper recipe	0.709	0.544	0.676
LLM, DistilGPT2 pretrained	0.621 ± 0.060	0.550	0.589 ± 0.038
LLM, GPT-2 from scratch	0.654 ± 0.012	0.538	0.544 ± 0.055

Reading Table 10. The language-model formulation trains end to end and produces valid per-event risk curves, clearly above chance. It trails our transformer by 0.06–0.09 on the simulator and METABRIC; on EBMT, where every model is near 0.54, it ties. The pretrained model is unstable across splits (sd up to 0.06). It is best presented as a proof of concept.

9 What the results support for the paper

Claim	Status	Evidence
Transformer with the $\mathcal{L}_{\text{PCH}}$ likelihood is competitive on tabular survival benchmarks	supported	best of the models we ran on METABRIC and SUPPORT; at the level of published PC-Hazard (Table 2)
...and matches or beats SurvTRACE	not supported	0.016–0.021 below published SurvTRACE
General multi-event formulation (non-, semi-, competing)	supported	best learned model on the simulator, hsa, EBMT; best true ordering; competing-risk DeepHit orders events worst (Tables 4, 5)
Neural model needed for non-linear risk	supported	Figure 1; hsa (Cox at chance)
$\mathcal{L}_{\text{rank}}$ improves prediction	not supported	neutral when tuned, harmful otherwise (Tables 3, 6)
$\mathcal{L}_{\text{mul}}$ improves within-patient ordering	not supported	worse on every proper true-order score (Table 5); gains only on a gameable metric
MMV improves predicted event times	partly supported	lower MAE on observed patients, worse calibration (Table 3)
Continuous numeric embedding is preferable to discretisation	partly supported	better on METABRIC, equal on SUPPORT (Table 7)
Coded sequences beat static summaries (§4.2)	not supported	Tables 8, 9
End-to-end LLM (§3)	proof of concept	works, weaker than the transformer (Table 10)

A further, methodological contribution follows from Section 4: the within-subject C-index rewards a loss trained on it beyond the oracle, so multi-event models should be evaluated with proper scores against true event order where it is available (simulation) and with this caveat elsewhere.

10 Limitations

- **Real multi-event data:** only EBMT, whose four categorical covariates carry little signal ($C_{td} \approx 0.54$ for every model); MIMIC requires credentialed access and the paper’s CKD/COPD/diabetes cohorts were not available.
- **Comparison with SurvTRACE** uses published numbers on different splits; our model has no multi-task head.
- **Number of splits:** 5 for benchmarks, 3 for sweeps and the LLM; small effects (< 0.005) are not resolvable.
- **MAE-margin is unreliable on EBMT** (values above 7000 days): with long follow-up and heavy censoring the Kaplan–Meier best guess extrapolates far beyond the data. MAE is reported only on METABRIC, SUPPORT and simulated data.
- **True-order scores** assume independent events given the covariates when converting marginal curves into $P(T_a < T_b)$; this is exact for the simulator’s non-competing regime and an approximation for semi-competing data.
- **§4.2 sequences** were tested only on simulated records with up to 256 tokens and $\approx 3\,000$ training patients; larger real cohorts or pre-trained clinical models may behave differently.

11 Open items

1. **Manuscript:** correct the sign of η in Eqs. 4–5 (the code is correct), add MENSA to Table 1 of the paper, present $\mathcal{L}_{\text{rank}}/\mathcal{L}_{\text{mul}}$ as optional penalties with this ablation, replace the within-subject

- C-index with true-order scores, and present the §3 language model as a proof of concept.
2. **§4.2:** decide between removing the claim, reporting the negative result, or testing on a real longitudinal cohort (MIMIC or the CKD/COPD/diabetes cohorts, loadable through `patients.csv` and `events.csv`), possibly with a pre-trained clinical encoder.
 3. **Gap to SurvTRACE:** add a multi-task head or compare on identical splits.
 4. **Real multi-event data with richer covariates** (MIMIC after credentialing).

A Hyper-parameters

Setting	single-event (METABRIC / SUPPORT)	multi-event, EHR
Transformer	tuned on validation over SurvTRACE’s grid: METABRIC 2 layers, hidden 16, intermediate 32, lr 10^{-3} SUPPORT 4 layers, hidden 16, intermediate 64, lr 10^{-4}	hidden 16, 2 layers, 4 heads, intermediate 16 (§4.2 “large”: hidden 64, 4 layers, intermediate 128)
Pooling	sum of layers, max over tokens	sum of layers, mean over tokens (“large”: [CLS])
Optimiser	AdamW, batch 64, weight decay 10^{-4} , ≤ 100 epochs	AdamW, lr 10^{-2} , weight decay 0.1, batch 256 (accum. 2), ≤ 200 epochs
Early stopping	10 evaluations without improvement on validation C_{td}	same
Time grid	4 intervals (quantiles 25/50/75 %)	10 intervals (quantiles of observed times)
Loss weights (fixed)	$\mathcal{L}_{PCH}$ 0.5, $\mathcal{L}_{rank}$ 0.2 ($\sigma = 0.1$)	$\mathcal{L}_{PCH}$ 0.5, $\mathcal{L}_{rank}$ 0.2, $\mathcal{L}_{mul}$ 0.1 ($\sigma = 0.1$)
MMV	scale balancing, $\lambda_v \in \{0.01, 0.1, 1\}$	$\lambda_v = 1$
LLM	–	30 epochs, early stopping 3; lr $5 \cdot 10^{-5}$ (pretrained), $5 \cdot 10^{-4}$ (scratch)

B Corrections made before the reported runs

All numbers in this report were produced after the following corrections; results obtained before them are superseded.

Issue	Effect and fix
Train/validation split swapped	models trained on $\approx 10\%$ of the data and early-stopped on $\approx 60\%$; fixed, with a regression test
Cox PH ignored categorical features	4 of 9 METABRIC and 6 of 14 SUPPORT features were constant; now one-hot encoded
SUPPORT numeric preprocessing	METABRIC’s column layout was hard-coded: five SUPPORT measurements unscaled, mean blood pressure lost; columns now follow each dataset’s layout (all SUPPORT results re-run in notebook 12)
“+ $\mathcal{L}_{mul}$ ” configuration	did not contain $\mathcal{L}_{mul}$; fixed
Checkpoint selection	used a C-index without τ -truncation and with IPCW fitted on the evaluated split; now selects on the SurvTRACE-protocol C_{td}
Error bars	seeds varied only the initialisation; each seed now uses its own split
Multi-event horizons	pooled across events, so late events were scored before they occurred; now per event
MENSA	an inspired re-implementation was reported under MENSA’s name; replaced by the published model
Prediction for new patients	numeric values never reached the model; fixed and verified against test predictions

C Where the numbers come from

All result files are in the repository under `results/new_results/<notebook>/`; every run has a `<tag>.json` with its metrics and a `<tag>.meta.json` with its full configuration.

Table / figure	notebook
Table 2, 3	03 (METABRIC), 12 (SUPPORT)
Table 4	01, 07 (re-runs)
Tables 5, 6	08
Figures 4–8	02
Figures 1–3	11
Table 7	12
Table 8	06, 09
Table 9, Figure 9	13
Table 10	05, 07